

Learning Activity Sheet for General Science

Quarter 2

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Learning Activity Sheet for General Science Grade 11
Quarter 2: Lesson 1

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LEARNING ACTIVITY SHEET

Learning Area:	General Science	Quarter:	2
Lesson No.:	1	Date:	
Lesson Title/ Topic:	Chemists and how they have made an impact on our lives		
Name:		Grade & Section:	11

Activity No. 1: Pasteur's Scientific Solutions

- I. Objective(s):** At the end of the activity, you should be able to apply Louis Pasteur's scientific discoveries to solve real-world health and safety challenges, demonstrating an understanding of germ theory, pasteurization, vaccination, and spontaneous generation. Through collaborative problem-solving and inquiry-based learning, you will:
- Evaluate how Pasteur's experiments transformed modern medicine, food safety, and microbiology.
 - Analyze and propose scientific solutions to health and sanitation-related problems.
 - Recognize the importance of scientific investigation in shaping public health practices and disease prevention.
 - Develop critical thinking skills while connecting historical scientific breakthroughs to modern applications.
- III. Materials Needed: Reference Materials** – Textbooks, online articles, or visual aids on Pasteur's discoveries (germ theory, pasteurization, vaccination, and spontaneous generation), Charts/Posters, Markers & Paper and presentation tools
- IV. Instructions:**
- Your team is a group of scientists assigned to solve mysterious disease outbreaks based on Louis Pasteur's discoveries. Read each scenario carefully and discuss the best scientific solution. Fill in your responses below.

1. Germ Theory Challenge – The Hospital Infection Mystery

Scenario

A hospital is struggling with patients developing infections after surgery. Some doctors believe diseases come from "bad air."

Your Task

Identify Pasteur's experiment that proved how germs cause infection. Describe the steps of the experiment and how it applies to hospital sanitation.

Response

2. Pasteurization Challenge – The Milk Safety Crisis**Scenario**

Farmers in a village complain that their milk spoils too quickly, making people sick.

Your Task

Explain Pasteur's process for preserving milk and preventing illness. Outline the scientific principle behind pasteurization.

Response

3. Rabies Vaccine Challenge – Emergency in the Barangay**Scenario**

A child was bitten by a rabid dog, but health workers don't know how to save him.

Your Task

Describe Pasteur's breakthrough in rabies vaccination. Explain how vaccines help the immune system fight diseases.

Response

4. Spontaneous Generation Challenge – The Bacteria Debate

Scenario

Some scientists argue that bacteria appear randomly in spoiled food.

Your Task

Identify Pasteur's experiment that disproved spontaneous generation.

Summarize the scientific method used to prove microorganisms come from living sources.

Response

Reflection Questions

1. How did Pasteur's discoveries change medicine and food safety?

2. If Pasteur had not conducted these experiments, how would the world be different today?

3. What modern scientific advancements build on Pasteur's work?
